Coefficient isolation and irrationality of sparse shifted-factorial series

Yuuki Miwa

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Abstract

We give a uniform coefficient-isolation lemma for multivariable series $\sum_{j\geq 0} z^{e_j}/(C_j - z^{e_j})$, where the exponent vectors give nonnegative integral coordinates for powers of an algebraic number $\lambda > 1$. A conjugate η with $0 < |\eta| < \lambda$, allowing $|\eta| \geq 1$, isolates a nonzero coefficient of $PF - T$ at a bounded weight, uniformly over all nonzero coefficients C_j and all $P \neq 0, T$ of bounded support. Combined with a weighted finite-model irrationality criterion, this proves irrationality of $\sum_n (\lfloor \lambda^n \rfloor! - 1)^{-1}$ for the positive roots of irreducible polynomials $X^d - \sum_{h<d} c_h X^h$ with nonnegative integer c_h, $c_0 > 0$, and $c_h > 0$ for at least one $h \geq 1$. Examples include the root of $X^3 - X - 1$ and a family of parameters tending to 1, none of whose positive integer powers is rational. The argument does not assume a stationary functional equation for the coefficients.

1 Statement and motivation

For $\lambda > 1$, choose an integer $n_0 \geq 0$ with $\lambda^{n_0} \geq 2$ and set

$$\xi_\lambda = \sum_{n \geq n_0} \frac{1}{\lfloor \lambda^n \rfloor! - 1}. \tag{1}$$

The series converges, and changing n_0 changes its value by a rational number. The shift by -1 removes the nested divisibility of ordinary factorial denominators. We study a class for which algebraic relations between the indices nevertheless yield useful finite rational models.

Theorem 1.1. *Let $d \geq 2$ and let*

$$p(X) = X^d - \sum_{h=0}^{d-1} c_h X^h \in \mathbb{Z}[X]$$

be irreducible over $\mathbb{Q}$, where $c_h \geq 0$, $c_0 \geq 1$, and $c_h > 0$ for at least one $h \geq 1$. Its unique positive root λ is greater than 1, and ξ_λ is irrational.

Corollary 1.2. *If $\theta > 1$ is the root of $X^3 - X - 1$, then*

$$\sum_{n \geq 3} \frac{1}{\lfloor \theta^n \rfloor! - 1} \notin \mathbb{Q}.$$

For each $d \geq 2$, the positive root λ_d of $X^d - 2X - 2$ also satisfies $\xi_{\lambda_d} \notin \mathbb{Q}$. Moreover, $\lambda_d \to 1$, and no positive integer power of λ_d is rational.

The elementary product-denominator argument illustrates the issue. Writing $L_n = \log(\lfloor \lambda^n \rfloor!)$, one has $L_{n+1}/L_n \to \lambda$. The logarithmic cost of clearing the initial segment through $s-1$ is asymptotic to $L_s/(\lambda - 1)$, while the first omitted term has size $\exp(-L_s)$.

Thus this first-order estimate gives irrationality when $\lambda > 2$, but does not give a vanishing integer error when $1 < \lambda < 2$. The proof below pays for a polynomial approximation to a block of terms instead of using just the initial segment.

The linear-algebra part is an elementary weighted Padé construction. The point requiring a separate argument is nonvanishing: small integer linear forms can be identically zero. Lemma 3.1 gives a coefficient bound uniform in an arbitrary sequence (C_j). A longer, fixed block then proves nonvanishing after evaluation, without increasing the denominator cost of the original construction. Section 6 compares this mechanism with related results. The unsparsified series $\sum_{n\geq 2}(n!-1)^{-1}$ is outside the argument.

2 A weighted finite-model criterion

Let $\beta_0, \dots, \beta_{d-1} > 0$ be linearly independent over $\mathbb{Q}$. For $a \in \mathbb{Z}_{\geq 0}^d$, write $w(a) = \sum_h a_h \beta_h$. The weighted order of a nonzero formal series is the minimum weight of a monomial with nonzero coefficient; set $\operatorname{ord}_w 0 = +\infty$. All bounded-weight sets are finite. Linear independence makes w injective on $\mathbb{Z}_{\geq 0}^d$, so each nonzero series has a unique least-weight monomial and $\operatorname{ord}_w(FG) = \operatorname{ord}_w F + \operatorname{ord}_w G$ for nonzero formal series F, G. The height $H(A)$ of an integer polynomial is the maximum of 2 and the absolute values of its coefficients. A polynomial family has fixed support if all its supports are contained in one fixed finite set.

Lemma 2.1. *Suppose $b_s \to \infty$ and $z_{s,h} > 0$ satisfy $z_{s,h} = b_s^{-\beta_h + o(1)}$. If $A_s \neq 0$ are integer polynomials of fixed support and $H(A_s) = b_s^{o(1)}$, and $\mu_s = \operatorname{ord}_w A_s$, then*

$$|A_s(z_s)| = b_s^{-\mu_s + o(1)}.$$

In particular, when $A_s(0) \neq 0$, this is a two-sided $b_s^{o(1)}$ estimate.

Proof. Distinct monomials in the fixed support have distinct weights. Their positive weight differences have a positive minimum. The unique least-weight coefficient has absolute value at least 1; every ratio of a higher-weight term to that term tends to zero, since coefficient heights and the errors in $z_{s,h}$ are subpower in b_s. There are only finitely many possible least monomials, so the estimate is uniform. □

Theorem 2.2. *Fix a nonempty finite set $\mathcal{A} \subset \mathbb{Z}_{\geq 0}^d$ and real numbers $R \geq \max_{a \in \mathcal{A}} w(a)$ and $\sigma > R$. Let $x \in \mathbb{R}$, $b_s \to \infty$, and $z_s \in \mathbb{Q}_{>0}^d$ satisfy $z_{s,h} = b_s^{-\beta_h + o(1)}$. Then x is irrational if the following conditions hold, with all supports, constants and thresholds fixed independently of s.*

(i) There are $S_s \in \mathbb{Q}$, $q_s \in \mathbb{Z}_{>0}$ and $\alpha \geq 0$ such that $q_s S_s \in \mathbb{Z}$ and $q_s \leq b_s^{\alpha + o(1)}$.

(ii) There are $\tau \in \mathbb{R}$ and $f_s = N_s/D_s$, where the integer polynomials N_s, D_s have fixed support and height $b_s^{o(1)}$, $D_s(0) \neq 0$, and $|x - S_s - f_s(z_s)| \leq b_s^{-\tau + o(1)}$.

(iii) There are integers $K_s > 0$ and $\kappa \geq 0$ such that $K_s z_s^a \in \mathbb{Z}$ for every $a \in \mathcal{A}$ and $K_s \leq b_s^{\kappa + o(1)}$.

(iv) The inequalities

$$2|\mathcal{A}| > \#\{c \in \mathbb{Z}_{\geq 0}^d : w(c) < \sigma\}, \qquad \alpha + \kappa < \min(\sigma, \tau) \tag{2}$$

hold.

(v) There are real numbers $J < \tau'$ and rational functions $g_s = \widetilde{N}_s/\widetilde{D}_s$ satisfying the same fixed-support and height conditions, with $\widetilde{D}_s(0) \neq 0$, such that

$$|x - S_s - g_s(z_s)| \leq b_s^{-\tau' + o(1)}.$$

For every s and every pair $P, T \in \mathbb{C}[z_0, \dots, z_{d-1}]$ supported in $\mathcal{A}$ with $P \neq 0$, one has $\operatorname{ord}_w(Pg_s - T) < J$.

Proof. Choose the coefficients of P_s, T_s supported in $\mathcal{A}$ to kill every coefficient of $E_s = P_sN_s - D_sT_s$ of weight less than σ. There are $2|\mathcal{A}|$ unknowns and fewer equations. The integer matrix has fixed size and height $b_s^{o(1)}$; maximal-rank minors give a nonzero integer kernel vector of height $b_s^{o(1)}$. Hence we may take $H(P_s), H(T_s) = b_s^{o(1)}$. If $P_s = 0$, the nonzero constant term of D_s gives $\operatorname{ord}_w(D_sT_s) = \operatorname{ord}_w T_s \leq R < \sigma$, a contradiction.

Set

$$U_s = q_sK_sP_s(z_s), \qquad V_s = (q_sS_s)K_sP_s(z_s) + q_sK_sT_s(z_s). \tag{3}$$

These numbers are integers by (i) and (iii). Lemma 2.1 applied to D_s, and the construction of E_s, give

$$|U_sx - V_s| \leq b_s^{\alpha+\kappa-\sigma+o(1)} + b_s^{\alpha+\kappa-\tau+o(1)} \longrightarrow 0. \tag{4}$$

For nonvanishing, use (v) for this same pair P_s, T_s. The integer polynomial $\widetilde{E}_s = P_s\widetilde{N}_s - \widetilde{D}_sT_s$ is nonzero and has least weight $\mu_s < J$, since multiplication by a denominator with nonzero constant term preserves order. Its support is fixed and its height is $b_s^{o(1)}$. Lemma 2.1 gives

$$|P_s(z_s)g_s(z_s) - T_s(z_s)| = b_s^{-\mu_s+o(1)}.$$

The remaining error, multiplied by $P_s(z_s)$, is at most $b_s^{-\tau'+o(1)}$. Since $\mu_s < J < \tau'$, it cannot cancel the displayed quantity for large s. Thus $0 < |U_sx - V_s| \to 0$. If x were rational, multiplication by its fixed denominator would give nonzero integers tending to zero. $\square$

Remark 2.3. The criterion separates the arithmetic elimination step from its nonvanishing input. In particular, the longer model in (v) does not enter the integer multiplier in (3).

3 Coefficient isolation using a conjugate

Let $\lambda > 1$ have algebraic degree d. Suppose that every power has nonnegative integral coordinates in its power basis:

$$\lambda^l = \sum_{h=0}^{d-1} e_{l,h}\lambda^h, \qquad e_l \in \mathbb{Z}_{\geq 0}^d \quad (l \geq 0). \tag{5}$$

Suppose also that a conjugate η satisfies

$$0 < |\eta| < \lambda, \qquad \rho = |\eta|/\lambda < 1. \tag{6}$$

Use the weights $\beta_h = \lambda^h$ and write $\mathcal{S}_R = \{a \in \mathbb{Z}_{\geq 0}^d : w(a) \leq R\}$. For arbitrary $C_l \in \mathbb{C} \setminus \{0\}$, define

$$F_C(z) = \sum_{l \geq 0} \frac{z^{e_l}}{C_l - z^{e_l}} = \sum_{l \geq 0}\sum_{r \geq 1} C_l^{-r} z^{re_l}. \tag{7}$$

This is a well-defined formal series: only finitely many pairs (l, r) can contribute below any prescribed weight, since $w(re_l) = r\lambda^l$.

Lemma 3.1. *Fix $R \geq 0$. Choose integers $j \geq 0$, $m \geq 1$ such that*

$$\lambda^j > R, \qquad m|\eta|^j(1-\rho) > 3R. \tag{8}$$

If $P \neq 0, T$ have support in $\mathcal{S}_R$, then for each nonzero coefficient p_a of P, the coefficient of z^{a+me_j} in $PF_C - T$ is exactly $p_aC_j^{-m} \neq 0$. Consequently, for any $J > R + m\lambda^j$,

$$\operatorname{ord}_w(PF_C - T) < J$$

uniformly over all the coefficients of P, T and all nonzero C_l.

Proof. A contribution to the indicated coefficient corresponds to

$$a + me_j = b + re_l, \qquad b \in \mathcal{S}_R, \quad r \geq 1, \quad l \geq 0. \tag{9}$$

Put $\Delta = b - a$, $\Delta_\lambda = \sum_h \Delta_h \lambda^h$ and $\Delta_\eta = \sum_h \Delta_h \eta^h$. Applying the real and conjugate embeddings to (9) and eliminating r yields

$$m(\eta^j - \lambda^j(\eta/\lambda)^l) = \Delta_\eta - \Delta_\lambda(\eta/\lambda)^l. \tag{10}$$

We have $|\Delta_\lambda| \leq R$ and $|\Delta_\eta| \leq 2R$, because $\left|\sum_h a_h \eta^h\right| \leq \sum_h a_h |\eta|^h \leq w(a) \leq R$, and the same holds for b. Thus the right side of (10) has modulus at most $3R$.

If $l \neq j$, the reverse triangle inequality gives

$$\begin{aligned} |\eta^j - \lambda^j(\eta/\lambda)^l| &= |\eta|^j |1 - (\eta/\lambda)^{l-j}| \\ &\geq |\eta|^j |1 - \rho^{l-j}| \geq |\eta|^j (1 - \rho). \end{aligned}$$

For negative $l - j$, the last inequality follows from $\rho^{-1} - 1 \geq 1 - \rho$; for positive $l - j$, it follows from $\rho^{l-j} \leq \rho$. This contradicts (8). Hence $l = j$, and then $|(m - r)\lambda^j| = |\Delta_\lambda| \leq R < \lambda^j$ forces $r = m$, and then the vector identity (9) gives $b = a$.

There is therefore exactly one contribution, $p_a C_j^{-m}$. Its weight is greater than R, so T contributes nothing there, and at most $R + m\lambda^j$, proving the order bound. □

Remark 3.2. The assumption is $|\eta| < \lambda$, not $|\eta| < 1$. The choices of j, m depend only on R, λ, η. They precede any choice of the coefficient sequence. Formal nonrationality for each fixed C alone would not supply this uniform order bound.

For example, the parameter λ_3 in Corollary 1.2 has minimal polynomial $X^3 - 2X - 2$, with discriminant -76 and $1 < \lambda_3 < 2$. Its two nonreal conjugates satisfy $|\eta|^2 = 2/\lambda_3 > 1$ and $|\eta| < \lambda_3$, since $\lambda_3^3 = 2\lambda_3 + 2 > 2$. Thus every nontrivial conjugate in this example lies outside the unit circle.

4 Factorial models and proof of the main theorem

We first prove irrationality under (5) and (6). All auxiliary parameters below are fixed before the limit $s \to \infty$ is taken.

Choice of supports. Put $A = 1/(\lambda - 1)$ and fix $1 < c < 2^{1/d}$. For large k, set $R = \lambda^k/2$ and $\sigma = cR$. Counting lattice points in a simplex gives

$$|\mathcal{S}_t| = \frac{t^d}{d! \prod_{h<d} \lambda^h} + O(t^{d-1} + 1).$$

Indeed, the union of the unit cubes based at $\mathcal{S}_t$ contains the simplex of weight at most t and is contained in the simplex of weight at most $t + \sum_h \lambda^h$. Since $c^d < 2$ and $1 < c < 2$, sufficiently large k gives

$$2|\mathcal{S}_R| > |\mathcal{S}_\sigma|, \qquad A + R < \sigma < \lambda^k. \tag{11}$$

Fix such k, R, σ.

Exact rational models. Set $X_s = \lambda^s$, $M_{s,h} = \lfloor \lambda^{s+h} \rfloor$, $b_{s,h} = M_{s,h}!$, $b_s = b_{s,0}$ and $z_{s,h} = b_{s,h}^{-1}$. For each fixed l, (5) gives

$$\lfloor \lambda^{s+l} \rfloor = \sum_h e_{l,h} M_{s,h} + r_{l,s}, \qquad 0 \le r_{l,s} < \sum_h e_{l,h}.$$

Multinomial integrality and factorial divisibility therefore give

$$\lfloor \lambda^{s+l} \rfloor ! = C_{l,s} \prod_h b_{s,h}^{e_{l,h}}, \qquad C_{l,s} \in \mathbb{Z}_{>0}. \tag{12}$$

For every fixed finite set of l, Stirling's formula implies

$$\log b_{s,h} = \lambda^h \log b_s + O(X_s), \qquad \log C_{l,s} = O(X_s), \qquad \log b_s \sim X_s \log X_s. \tag{13}$$

The leading terms in the factorial quotient cancel by (5). Thus $z_{s,h} = b_s^{-\lambda^h + o(1)}$ and every fixed block

$$f_{s,t}(z) = \sum_{l<t} \frac{z^{e_l}}{C_{l,s} - z^{e_l}} \tag{14}$$

has an integer numerator and denominator of fixed support and height $b_s^{o(1)}$. The denominator has constant term $\prod_{l<t} C_{l,s} \neq 0$. At z_s, (14) equals exactly the terms of (1) with $s \le n < s+t$.

Initial segment, tails, and denominator cost. Let S_s be the initial segment with $n < s$, and take

$$q_s = \prod_{n_0 \le n < s} (\lfloor \lambda^n \rfloor ! - 1).$$

Then $q_s S_s \in \mathbb{Z}$. From $L_{n+1}/L_n \to \lambda$ and a geometric-sum comparison, $\sum_{n<s} L_n/L_s \to A$. Consecutive terms of the series have ratio tending to zero. Hence, for each fixed t,

$$q_s = b_s^{A+o(1)}, \qquad \xi_\lambda - S_s - f_{s,t}(z_s) = b_s^{-\lambda^t + o(1)}. \tag{15}$$

The latter equality is a two-sided estimate of a positive tail.

For the support $\mathcal{S}_R$, define

$$W_s = \max_{a \in \mathcal{S}_R} \sum_h a_h M_{s,h}, \qquad K_s = W_s!.$$

For every supported monomial,

$$\prod_h (M_{s,h}!)^{a_h} \mid \left(\sum_h a_h M_{s,h} \right)! \mid W_s!.$$

Moreover $W_s \le R X_s$, so $K_s \le b_s^{R+o(1)}$. Thus the first four conditions of Theorem 2.2 hold for $f_s = f_{s,k}$ with $\alpha = A$, $\kappa = R$, and $\tau = \lambda^k$.

The nonvanishing input. Choose j, m, J as in Lemma 3.1, and then a fixed $\ell \ge k$ with $\lambda^\ell > J$. Apply the lemma to the actual full sequence $(C_{l,s})_{l \ge 0}$. Terms of its formal series with $l \ge \ell$ start at weight $\lambda^\ell > J$. Therefore the same nonzero coefficient below weight J occurs in $Pf_{s,\ell} - T$ for every supported $P \neq 0, T$. The fixed long block $g_s = f_{s,\ell}$ satisfies the height conditions by (13), and the tail estimate with $\tau' = \lambda^\ell$ by (15). Condition (v) follows. Theorem 2.2 proves irrationality under (5)–(6).

Completion of the proof of Theorem 1.1. Descartes' rule of signs and $p(1) < 0$ show that p has exactly one positive root, and it exceeds 1. Irreducibility makes its degree d. Starting with the standard basis vectors $e_0, \ldots, e_{d-1}$, the recurrence $e_{l+d} = \sum_{h<d} c_h e_{l+h}$ gives nonnegative integral vectors satisfying (5).

The product of the moduli of all conjugates is c_0, whereas $\lambda^d = c_0 + \sum_{h\ge 1} c_h \lambda^h > c_0$. Thus some other conjugate has modulus less than λ; otherwise the product would be at least λ^d. It is nonzero since $c_0 \neq 0$. This proves (6) and completes the argument. □

Proof of Corollary 1.2. The cubic has no rational root, so is irreducible. Its root satisfies $\theta^2 < 2 < \theta^3$, allowing $n_0 = 3$. Each $X^d - 2X - 2$ is irreducible by Eisenstein's criterion at 2. Its positive root satisfies $1 < \lambda_d < 3$ and $\lambda_d^d = 2\lambda_d + 2 < 8$, whence $0 < \log \lambda_d < (\log 8)/d \to 0$. Finally, if $\lambda_d^q \in \mathbb{Q}$ for some positive integer q, every conjugate ζ would satisfy $\zeta^q = \lambda_d^q$ and thus $|\zeta| = \lambda_d$. This contradicts the smaller-modulus conjugate established above. □

5 An exact certificate for the cubic example

This section is optional for the proof of the theorem, but makes the fixed parameters concrete. The values below verify Theorem 2.2 directly; they need not follow the asymptotic choice $R = \lambda^k/2$ in Section 4. For the root θ of $X^3 - X - 1$, $13/10 < \theta < 4/3$. Its nonreal conjugates have modulus $\theta^{-1/2}$, so $\rho = \theta^{-3/2}$ and $\rho^2 = 1/(\theta + 1)$. The recurrence gives

$$\theta^{11} = 4 + 7\theta + 5\theta^2, \qquad \theta^{12} = 5 + 9\theta + 7\theta^2.$$

One may use

$$k = 12, \quad R = 20, \quad \sigma = 24, \qquad j = 11, \quad m = 1000, \quad J = 23000, \quad \ell = 36.$$

Indeed, $20 < \theta^{11} < 200/9$, so $|\eta|^{11} > 3/\sqrt{200} > 21/100$. Also $\rho^2 < 10/23 < 49/100$, and consequently

$$1000|\eta|^{11}(1-\rho) > 1000 \cdot \tfrac{21}{100} \cdot \tfrac{3}{10} = 63 > 60 = 3R.$$

The isolated coefficient has weight less than $20 + 200000/9 < 23000$. Further,

$$\theta^{12} > \frac{2853}{100} > \frac{57}{2}, \qquad \theta^{36} > \left(\frac{57}{2}\right)^3 = \frac{185193}{8} > 23000,$$

and $20 + 1/(\theta - 1) < 20 + 10/3 < 24 < \theta^{12}$. Exact rational interval enumeration gives $|\mathcal{S}_{20}| = 766$ and $|\mathcal{S}_{24}| = 1264 < 1532$. The accompanying script checks these counts, the power-vector identities and the displayed rational inequalities. The proof for all coefficients is Lemma 3.1; finite enumeration is not its substitute.

6 Relation to existing methods and scope

The general criterion packages finite linear elimination, arithmetic height bounds and the elementary irrationality principle $0 < |Ux - V| \to 0$. Its application-specific input is the nonvanishing estimate, rather than a new linear-algebra principle.

Mingarelli's abstract-factorial theorem proves irrationality for reciprocal series whose denominators satisfy specified binomial-integrality and factorial-divisibility axioms [8, Definition 1 and Theorem 53]. Directly substituting the denominators of (1) does not meet those axioms: they are odd, whereas an abstract factorial at index $n \ge 2$ must be divisible by $n!$. A reformulation would need an additional argument. The multinomial divisibility used in (12) concerns the unshifted factorials inside a rational model.

Erdős's general criterion requires $\limsup a_n^{1/2^n} = \infty$, together with a polynomial lower bound [4, Theorem 1]. For $a_n = \lfloor \lambda^n \rfloor! - 1$,

$$\log a_n \sim n\lambda^n \log \lambda,$$

so its main growth condition fails when $1 < \lambda < 2$. Hančl and Luca's rational-function criterion [5, Theorem 2] applies to our series with $a_n = \lfloor \lambda^n \rfloor!$, $K = \mathbb{Q}$, $R = 1$, $P_1 = 1$ and $Q_1 = X - 1$. Its degree parameter is $d = 1$, so it requires $\limsup a_n^{1/2^n} = \infty$. Since

$$\frac{\log a_n}{2^n} \sim n \log \lambda \left(\frac{\lambda}{2}\right)^n,$$

the growth condition holds exactly when $\lambda \geq 2$, where the criterion proves irrationality. The remaining hypotheses hold here, including the stronger condition $\log a_n / \log n \to \infty$. Their Corollary 4 with $\alpha = -1$ gives the same conclusion. Neither substitution covers the plastic constant or the parameters of Corollary 1.2 below 2. Hančl and Tijdeman's factorial-series criteria use integer numerators [6, Lemma 2.2]. They apply to the unshifted sparse factorial sum, but the numerator $m!/(m! - 1)$ obtained by rewriting a shifted term over $m!$ is not an integer for $m \geq 3$. Barreto et al. obtain growth thresholds tending to 1 for overlapping products of consecutive denominator terms [2, Theorems 2–3]. That product representation is not supplied by our shifted denominators; the one-factor specialization has threshold 2.

There is a useful reason to retain the exact denominator structure. Kovač and Tao show that a strictly increasing positive integer sequence (a_n) with $\sum_n 1/a_n < \infty$ and $a_{n+1}/a_n^2 \to 0$ admits positive integer perturbations $b_n \sim a_n$ whose reciprocal sum is rational [7, Theorem 2.4]. Our denominators satisfy this condition when $1 < \lambda < 2$. Thus their coarse asymptotics alone cannot force irrationality. This observation concerns nearby sequences, not the value of (1) itself.

Zero-order estimates for Mahler functions are a close methodological comparison. For a nonrational Mahler function, Coons's rational-approximation estimate bounds the contact order using the order and polynomial degrees of its equation [3, Lemma 1]; uniformly bounded equation data could therefore give a family-uniform bound. Multivariable Mahler theory likewise treats finite systems under suitable admissibility assumptions [1, Definition 2.1 and Theorem 2.3]. The immediate identity for our formal model is

$$F_C(z) = \frac{z^{e_0}}{C_0 - z^{e_0}} + F_{\mathrm{shift}(C)}(\mathcal{M}z),$$

where $(\mathcal{M}z)_h = z^{e_{h+1}}$, so substitution sends z^{e_l} to $z^{e_{l+1}}$. For arbitrary C, this gives an infinite succession of functions. No finite Mahler system with uniformly bounded data is established for it here.

Töpfer's generalized zero estimate [9, Theorem 1] bounds nonzero polynomial evaluations. It allows formal coefficients in characteristic zero and a rational transformation, but requires a finite univariate functional system; the constant also depends on the vanishing order of its denominator after substitution. His more abstract Theorem 2 permits varying auxiliary polynomial families, subject to quantitative hypotheses, including a bound on the valuative proximity of a fixed tuple to every common zero. These hypotheses have not been verified for our family. Thus nonstationarity alone is not a reason to exclude that theorem. Lemma 3.1 gives an explicit coefficient identity without those auxiliary inputs.

The input used here is the explicit bound in Lemma 3.1, uniform over the entire coefficient family. Its application to a fixed longer model supplies nonvanishing without increasing the integer denominator cost in Theorem 2.2.

The parameters in Theorem 1.1 are irrational algebraic integers; rational parameters require separate arguments. The theorem does not cover every real $\lambda > 1$ or every algebraic integer. The nonnegative power coordinates and a nonzero conjugate of smaller modulus are sufficient hypotheses. No transcendence, irrationality-exponent, signed-perturbation, or unsparsified-series conclusion is asserted here.

Supplementary verification. The finite certificate in Section 5 is checked by an accompanying Python program using exact integer and rational arithmetic. The file `verify_conjugate_isolation.py` and its JSON output accompany this paper. The proof is self-contained and does not depend on numerical solution of a Padé system or on a computer search.

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